CSA1506 - Cloud Computing and Big Data Analysis

B. Praveen Kumar
192511135

CO4 - Assessment Tool-1

Class Test

1. Unified Big Data Architecture for Retail:

- Retail Organization can integrate all these sources into central data lake
- Retail organization collect data from websites, Mobile apps, stores, POS system
- Kafka can be used for real time data integration
- Spark can process large scale batch and streaming batch
- Machine learning can be analyse customer behaviour and purchase history.
- The system can provide personalized recommendation offers and promotion

2. Distributed Computing for Specific Research

- Scientific research produces very large dataset like that are difficult for centralized systems to process
- Distributed computing divide data and tasks among multiple computers.

Parallel processing allows several tasks to execute at the same time.

Technologies such as Hadoop and spark support distributed processing

It provides faster processing, scalability, and better resource ultilization.

3. Fraud Detection in digital Payments;

~~Fra~~ Digital payment system generates a continuous stream of transaction.

Kafka can collect transaction data in every real time

Spark Streaming can analyse transaction immediatly

Machine learning models can identify unusal transaction patterns.

Parameter such as amount, location, frequency device and transaction time can be analyzed.

4. Multimedia Data in Big Data:

Multimedia include images, audio, video, which requires large storage capacity

Major challenges include ~~sto~~ large size, unstructured format, processing complexity, and high bandwidth requirements.

Cloud object storage such as Amazon S3 or similar services can store multimedia data

HDFS can provide distributed storage

Sparks can process distributed storage.

5. Disaster Recovery and Business Continuity

Big data platform must continue operating during hardware, network, or system failures.

Redundancy maintains multiple system components

Replication creates copies of important data

Regular backups protect against data loss

Automated failover switches to a backup system when the primary system fails

Disaster recovery sites can be maintained in different location.

6. Data Quality and Data Governance:

Poor data quality includes inconsistency duplication missing values and incorrect data

It can produce inaccurate analysis and poor business decisions

Data cleaning removing incorrect and duplicate records

Validation rules can identify missing or invalid values.

Data standards ensure consistency across different systems.

Data governance defined ownership, security, quality and access policies.

7. Recommendations System for Online streaming:

Streaming platform do collect watch History, ratings, searches, likes and viewing duration

This data can be stored in distributed data platform

Kafka can collect real time user activity

Spark can process large datasets

Recommendation algorithms such as collaborative filtering and content based filtering can be used

This increases user engagement and customer retention

8. Cloud computing in Big Data:

Advantages:

Provides high scalability

Support pay-as-you-use-pricing

Reduces hardware and maintainance cost

provides flexible storage and computing resources

Limitations:

Security and privacy concerns

Dependence on internet connectivity

Possible vendor lock in

cloud cost can increase if not monitored properly

9. Real Time Transportation Analytics

Transportation systems generate data from GPS sensor, traffic system and vehicles.

Real-time data can be collected using streaming technologies such as Kafka

Spark can process the data continuosly

Analytical can identify traffic and optimize routes

This helps reduce fuel consumption travel time and delivery delays.

10 Ethical and Legal Challenges of Big Data

Big Data can create privacy, security, bias and data misuse issues.

Personal information should not be collected or used without proper consent

Organizations should follow applicable privacy and data protection laws

Data should be protected using encryption and access control

Only necessary data should be collected

Organizations should provide transparency about data usage.